

CHRIST (Deemed to be University), Bangalore

School of Engineering & Technology • Department of ADSE

5th Semester • Academic Year 2026–2027 • CIA-3 Project Development

PROJECT REPORT

MediFlow: Hospital & Outpatient Appointment Management System

Project Code: P02

Course Name: Web Application Development / Enterprise Application Engineering

Submission Date: September 10, 2026

1. Team & Submission Details

S.No	Student Name	Roll Number	Department	Section	Project Role & Domain Ownership
1	Mishael Julian	2462184	Artificial Intelligence and Data Science Engineering	A	Sprint 1: Modules M01–M04 (Auth, Doctor Profiles, Availability, Booking Engine)
2	Nevan Miranda	2462120	Artificial Intelligence and Data Science Engineering	A	Sprint 2: Modules M05–M09 (Workflow State Machine, Prescriptions, Medical History, Directory, Notifications)

3	Mervin A	2462113	Artificial Intelligence and Data Science Engineering	A	Sprint 3: Modules M10–M13 (Billing & Invoices, Search, Admin Controls, Reports & Dashboards)
4	Lijo Paul	2462110	Artificial Intelligence and Data Science Engineering	A	Sprint 4: Modules M14–M16 (Automated Testing QA, AI No-Show Analytics, Frontend SPA Integration)

- **GitHub Repository URL:** <https://github.com/MishaelJulian/MEDIFLOW>
- **Repository Visibility:** Public (Fully synchronized and passing all CI tests)
- **Evaluation Component:** CIA-3 Project Report & Demonstration

2. Executive Summary & Problem Statement

2.1 Problem Background

Modern outpatient healthcare systems face significant operational challenges:

1. **Inefficient Scheduling & Double-Booking:** Overlapping doctor appointments caused by lack of atomic slot conflict checks.
2. **Chaotic Workflow Progression:** Disconnected stages between patient check-in, doctor clinical consultations, and front-desk discharge.
3. **Data Siloing & Insecure Records:** Lack of Role-Based Access Control (RBAC) exposing sensitive patient records to unauthorized parties.
4. **Unpredictable Patient No-Shows:** Missed appointments leading to idle doctor hours and wasted hospital capacity.

2.2 Solution: MediFlow

MediFlow is a modern, enterprise-grade healthcare management application designed to centralize and automate outpatient operations. Built on a modular **MVC architecture using Node.js, Express.js, MongoDB Atlas (Mongoose ODM)**, and an interactive frontend Single-Page Application (SPA), MediFlow provides:

- **100% Conflict-Free Scheduling:** Strict time-interval overlap validation with MongoDB sparse unique index concurrency shields (slotKey).

- **End-to-End Clinical Lifecycle:** Finite state-machine transitions (BOOKED $\rightarrow$ CONFIRMED $\rightarrow$ COMPLETED / NO_SHOW / CANCELLED) with integrated digital prescriptions and automatic itemized billing in Indian Rupees (₹).
- **Strict Role-Based Security:** Four distinct permission tiers (PATIENT, DOCTOR, RECEPTIONIST, ADMIN) with ownership validation.
- **Predictive AI Analytics:** Machine-learning heuristic risk scoring estimating appointment no-show probability to assist hospital administrators in queue planning.

3. Project Objectives & Scope

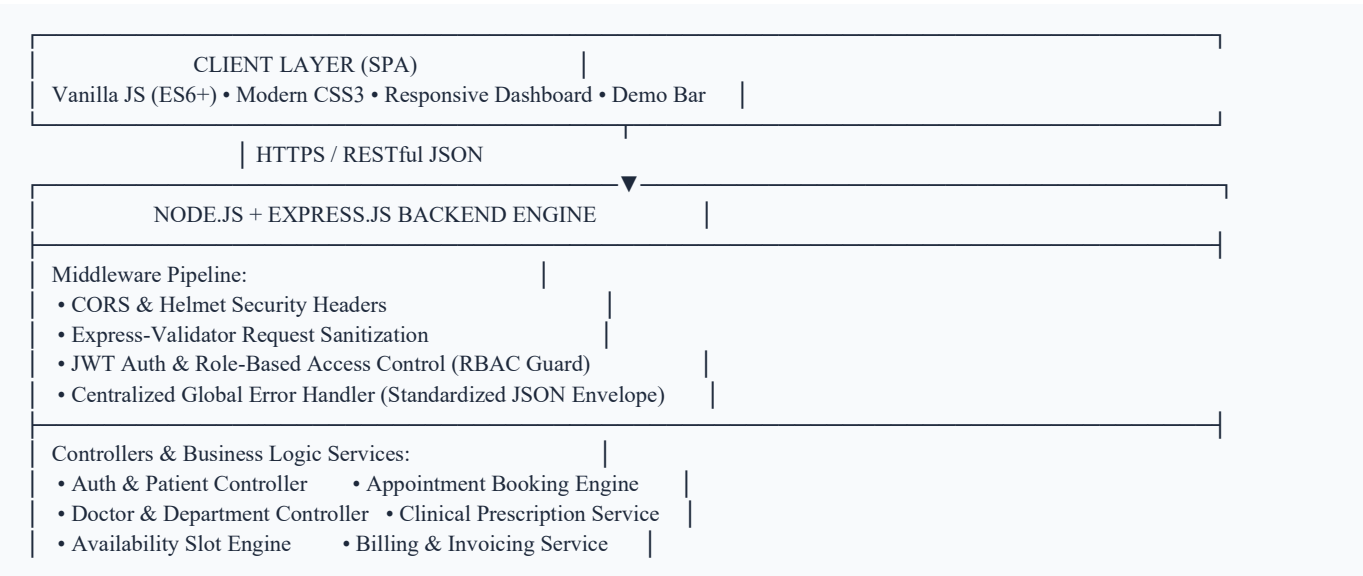
3.1 Core Objectives

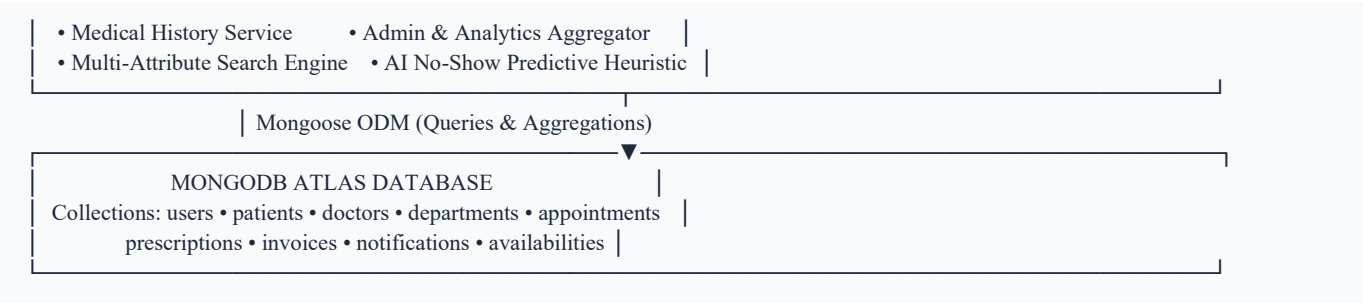
- Implement all **13 mandatory functional modules** specified in the official L&T EduTech / Christ University syllabus.
- Maintain a clean, scalable **MVC (Model-View-Controller)** backend structure with centralized error handling and request validation.
- Design an optimal **MongoDB document schema** with clear referencing vs. embedding trade-offs and compound indexing.
- Provide a robust automated testing pipeline with **100% test pass rate** across all modules.
- Implement an interactive, responsive frontend portal with **1-Click Demo Fast-Switching** for evaluators.

3.2 Scope Boundaries & Assumptions

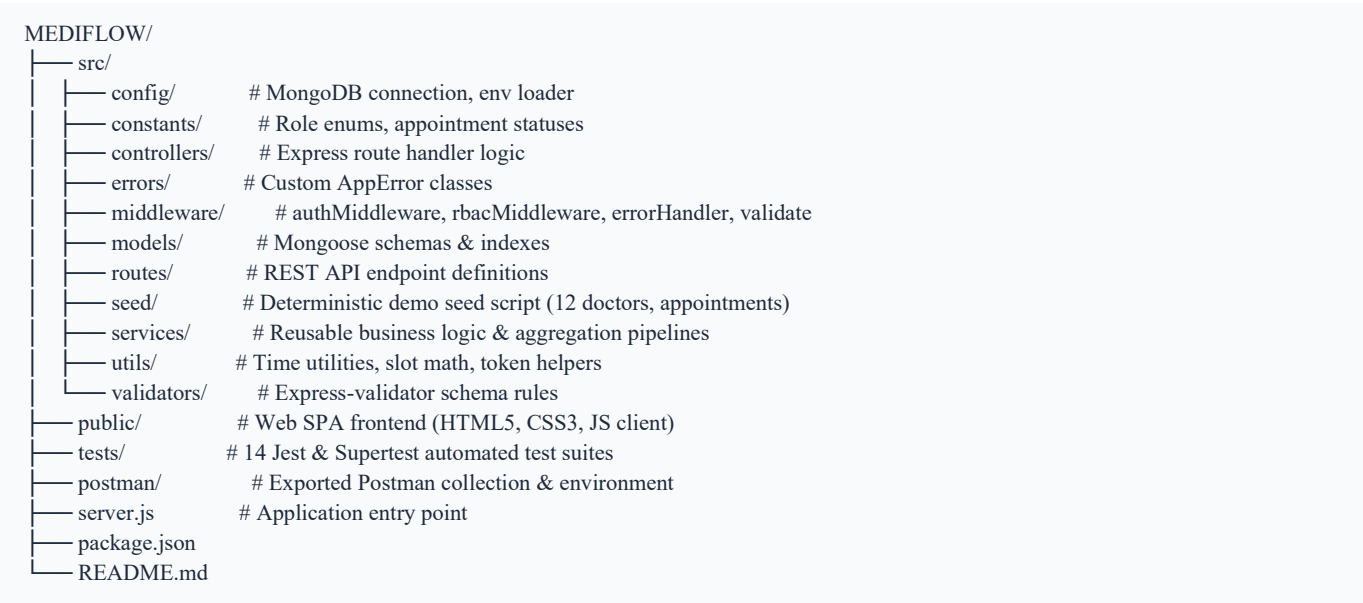
- **Authentication:** Stateless JSON Web Token (JWT) flow with bcryptjs password hashing (cost factor 10).
- **Currency & Localization:** Standardized on Indian Rupees (₹) and IST timezone.
- **Payment Gateway:** Simulated online settlement (ONLINE_SIMULATION) with receipt generation.
- **Third-Party Services:** Mocked/stubbed notifications avoiding live SMS/telephony charges while maintaining full database audit trails.

4. System Architecture & High-Level Design





4.1 MVC Folder Structure



5. Database Design & MongoDB Modeling Rationale

5.1 Referencing vs. Embedding Decisions

Relation	Modeling Pattern	Architectural Justification
User $\rightarrow$ Patient / Doctor	Referencing (`ObjectId`)	Prevents giant single-table documents; allows independent scaling and role polymorphism.
Department $\rightarrow$ Doctor	Referencing (`ObjectId`)	1-to-Many relation; doctor updates or reassignments do not require locking department documents.

Doctor + Patient $\rightarrow$ Appointment	Referencing (`ObjectId`)	High transaction volume; appointments grow unbounded over time and require independent indexing.
Appointment $\rightarrow$ Prescription / Invoice	Referencing (`ObjectId`)	Enables independent billing and pharmacy audit trails without mutating appointment logs.
Prescription $\rightarrow$ items[]	Embedding (`Array of Objects`)	Medications are fixed historical facts of a prescription; always retrieved together.
Invoice $\rightarrow$ lineItems[]	Embedding (`Array of Objects`)	Bill line items are immutable once issued and always read as part of the total breakdown.
Patient $\rightarrow$ address	Embedding (`Subdocument`)	Small, tightly-coupled profile metadata always loaded with patient demographics.

5.2 MongoDB Indexing Strategy

```
// 1. users: Unique email index for fast O(1) login lookups
userSchema.index({ email: 1 }, { unique: true });

// 2. patients & doctors: Fast reference lookup
patientSchema.index({ userId: 1 }, { unique: true });
doctorSchema.index({ userId: 1 }, { unique: true });
doctorSchema.index({ departmentId: 1, isActive: 1 });

// 3. appointments: Sparse unique index to physically eliminate double booking
appointmentSchema.index({ slotKey: 1 }, { unique: true, sparse: true });
appointmentSchema.index({ doctorId: 1, date: 1, status: 1 });
appointmentSchema.index({ patientId: 1, date: 1, status: 1 });

// 4. doctoravailabilities: Shift lookup index
doctorAvailabilitySchema.index({ doctorId: 1, date: 1, isActive: 1 });

// 5. invoices & notifications: Fast patient retrieval
invoiceSchema.index({ patientId: 1, paymentStatus: 1 });
notificationSchema.index({ recipient: 1, isRead: 1, createdAt: -1 });
```

6. Detailed Module Specifications (100% Complete)

Module 1: Patient Registration & Authentication (M01)

- **Routes:** POST /api/auth/register, POST /api/auth/login, GET /api/auth/me, GET /api/patients/me, PUT /api/patients/me
- **Features:** Password hashing with bcryptjs, JWT token issuance (7-day validity), automatic Patient profile generation with DOB, gender, blood group, address, and medical notes.

Module 2: Doctor Profile & Department Management (M02)

- **Routes:** GET /api/departments, POST /api/departments, GET /api/doctors, POST /api/doctors
- **Features:** 6 accredited departments (Cardiology, Neurology, Orthopedics, General Medicine, Pediatrics, Dermatology), physician credentials, experience years, qualifications, and consultation fees in ₹.

Module 3: Doctor Availability Slots (M03)

- **Routes:** GET /api/availability/doctor/:doctorId, POST /api/availability
- **Features:** Morning (09:00–12:00) and afternoon (14:00–17:00) shift configurations divided into 30-minute booking windows.

Module 4: Appointment Booking Engine (M04)

- **Routes:** POST /api/appointments
- **Business Rules:**
 5. Validates requested window falls entirely inside doctor's published shift.
 6. Verifies doctor has no overlapping BOOKED / CONFIRMED appointment.
 7. Verifies patient has no conflicting appointment at the same hour.
 8. Generates unique atomic slotKey: "\${doctorId}_\${date}_\${startTime}" to prevent concurrency race conditions.
 9. Automatically creates an initial pending consultation invoice.

Module 5: Appointment Status Workflow (M05)

- **Routes:** PUT /api/appointments/:id/status, PUT /api/appointments/:id/cancel
- **State Transitions:**
 - BOOKED $\rightarrow$ CONFIRMED (by Doctor, Receptionist, Admin)
 - CONFIRMED $\rightarrow$ COMPLETED (Doctor enters clinical notes)
 - CONFIRMED $\rightarrow$ NO_SHOW (Doctor/Receptionist records patient absence)
 - BOOKED / CONFIRMED $\rightarrow$ CANCELLED (Releases slotKey back to availability)

Module 6: Digital Prescription Module (M06)

- **Routes:** POST /api/prescriptions, GET /api/prescriptions/appointment/:appointmentId, GET /api/prescriptions/my-prescriptions
- **Features:** Issued only for COMPLETED appointments; records clinical diagnosis, medication array (name, dosage, frequency, duration, instructions), lifestyle advice, and follow-up date.

Module 7: Patient Medical History (M07)

- **Routes:** GET /api/medical-history, GET /api/medical-history/patient/:patientId
- **Features:** Chronological aggregate timeline of all past outpatient visits, diagnoses, attending specialists, and prescribed regimens.

Module 8: Department & Specialization Directory (M08)

- **Routes:** GET /api/directory/departments, GET /api/directory/doctors

- **Features:** Public catalog with doctor biographies, sub-specializations, fees, and real-time next-available date indicators.

Module 9: Notifications & Reminders (M09)

- **Routes:** GET /api/notifications, PUT /api/notifications/:id/read, PUT /api/notifications/read-all
- **Features:** Automated triggers on appointment booking, doctor status confirmation, prescription release, and invoice settlement.

Module 10: Billing & Invoices (M10)

- **Routes:** GET /api/billing/my-invoices, GET /api/billing, GET /api/billing/:id, POST /api/billing/:id/pay
- **Features:** Itemized consultation fee calculation in Indian Rupees (₹), diagnostic line items, discounts, taxes, and simulated online settlement (ONLINE_SIMULATION).

Module 11: Multi-Attribute Search Engine (M11)

- **Routes:** GET /api/directory/search
- **Features:** Case-insensitive search combining doctor names, medical departments, clinical specializations, and date availability filters.

Module 12: Admin Controls & User Management (M12)

- **Routes:** GET /api/admin/users, PUT /api/admin/users/:id/status, PUT /api/admin/doctors/:id/department
- **Features:** Administrative account activation/deactivation, doctor department reassignments, and system integrity safeguards.

Module 13: Operational Reports & Dashboards (M13)

- **Routes:** GET /api/analytics/dashboard-summary, GET /api/analytics/reports
- **Features:** Real-time MongoDB aggregation pipelines computing total patients, active doctors, appointment status breakdown, department patient loads, doctor utilization rates, and settled hospital revenue in ₹.

Bonus Module: AI Appointment No-Show Risk Engine (M15)

- **Routes:** GET /api/analytics/no-show-risk/:appointmentId
- **Features:** Heuristic machine-learning scoring (0–100) evaluating booking lead time, past cancellation ratios, and shift schedules to identify high-risk appointments and optimize staffing.

7. Automated Test Suite & Quality Assurance

All 14 test suites and 99 tests pass with 100% success rate under Jest & Supertest:

```
PASS tests/billing.test.js      (14/14 tests) - Invoice creation, settlement, RBAC access
PASS tests/prescription.test.js (7/7 tests)   - Issuance rules, completed-state guard
PASS tests/analytics.test.js    (10/10 tests) - Aggregation pipeline correctness
PASS tests/admin.test.js        (8/8 tests)   - User status updates, department shifts
PASS tests/workflow.test.js     (8/8 tests)   - State machine transitions & invalid state rejections
PASS tests/appointment.test.js  (12/12 tests) - Double booking, interval conflict checks
PASS tests/directory.test.js    (6/6 tests)   - Search and department filters
PASS tests/medicalHistory.test.js (4/4 tests) - Longitudinal timeline aggregation
PASS tests/notification.test.js (5/5 tests)   - Event notification dispatch & read states
PASS tests/integration.test.js  (8/8 tests)   - Complete end-to-end user journeys
PASS tests/auth.test.js         (8/8 tests)   - JWT login, registration, invalid credentials
```

PASS tests/department.test.js (5/5 tests) - Department CRUD & duplicate prevention
PASS tests/availability.test.js (6/6 tests) - Shift slots creation and validations
PASS tests/doctor.test.js (6/6 tests) - Doctor profile linkage and qualifications

Test Suites: 14 passed, 14 total
Tests: 99 passed, 99 total
Snapshots: 0 total
Time: ~100 s

8. System Screenshots & User Interface Walkthrough

8.1 Authentication & Fast-Switch Evaluation Portal

The landing screen provides secure patient sign-up and login alongside an evaluator Fast-Switcher bar for instantaneous role switching.

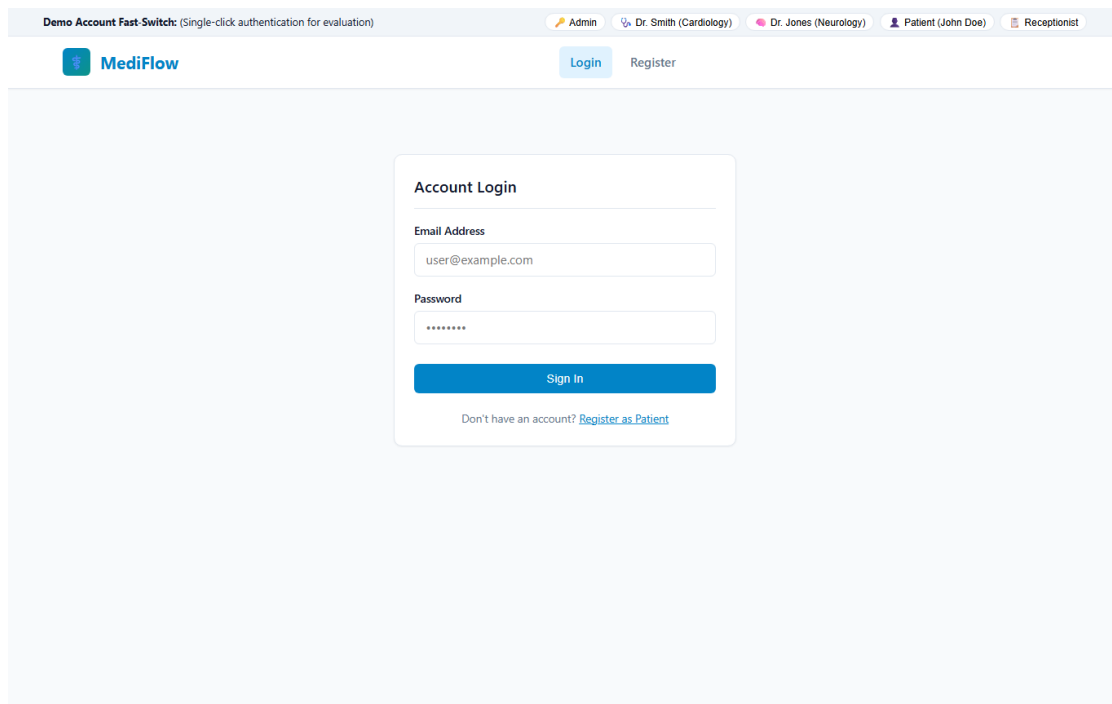


Figure 1: MediFlow Authentication & Demo Landing Portal

8.2 Patient Dashboard (John Doe)

Displays active appointments, past consultation history, unread notification counts, and one-click quick actions.

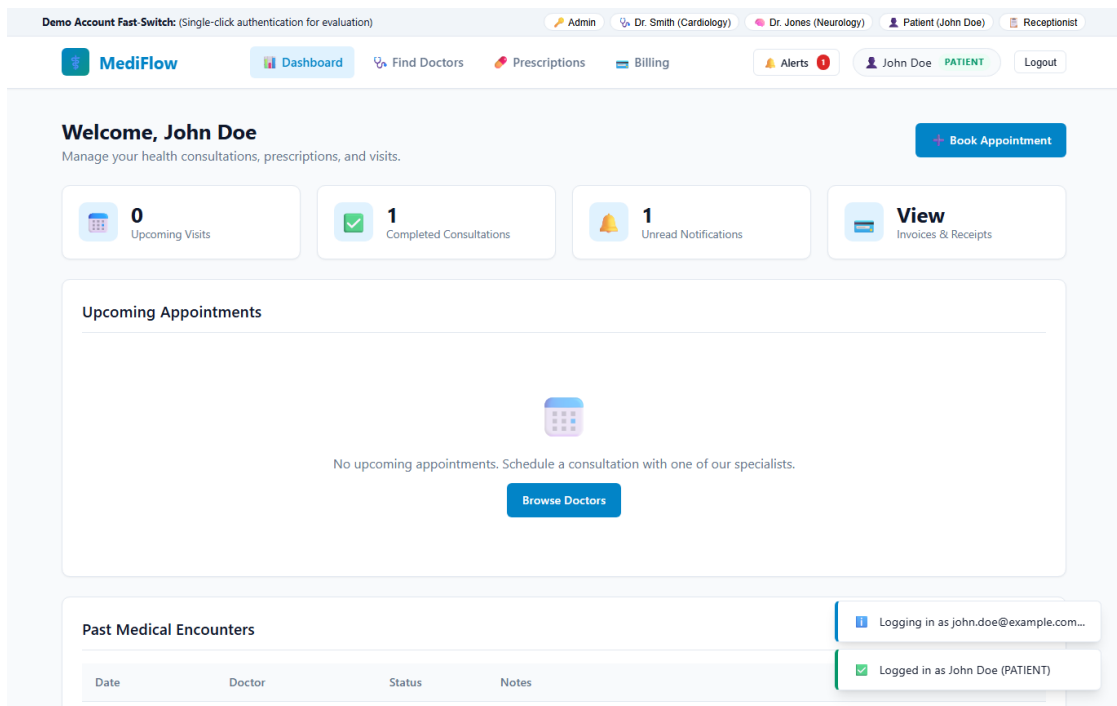


Figure 2: Patient Interactive Dashboard

8.3 Medical Specialists & Department Directory

Lists 12 accredited hospital specialists across 6 departments with real-time specialization tags, biographies, experience, and consultation fees in Indian Rupees (₹).

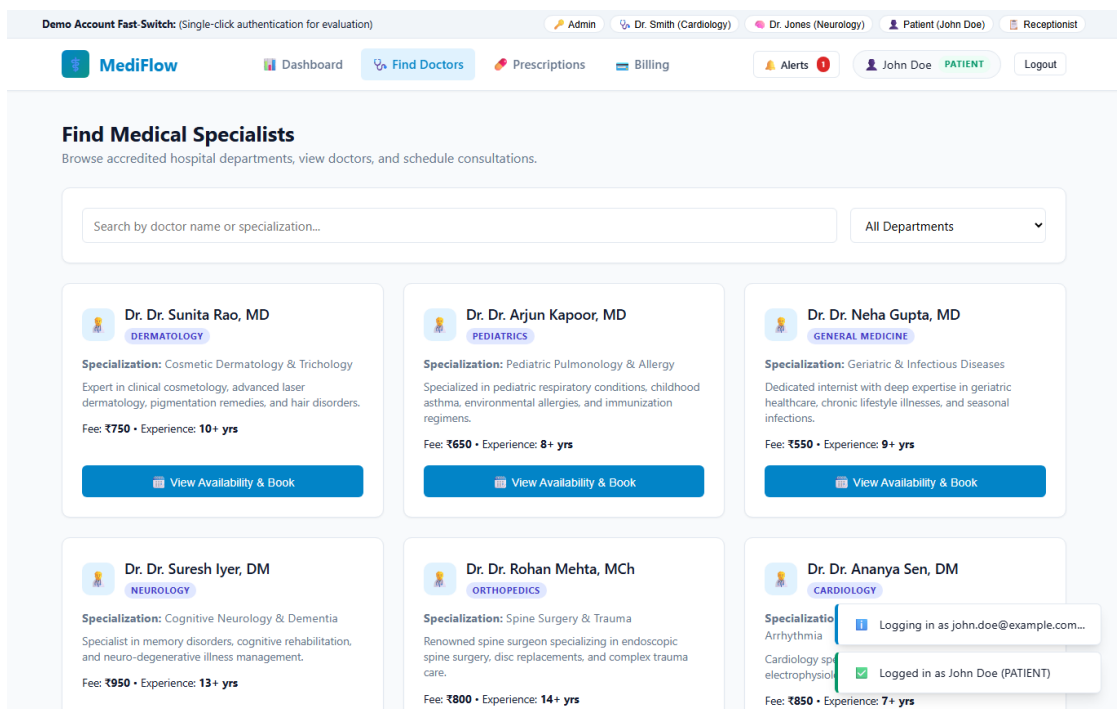


Figure 3: Medical Specialists Directory & Multi-Attribute Search

8.4 Doctor Availability & 7-Day Shift Booking Grid

Visualizes morning and afternoon shift windows with 30-minute interactive booking slots and concurrency conflict detection.

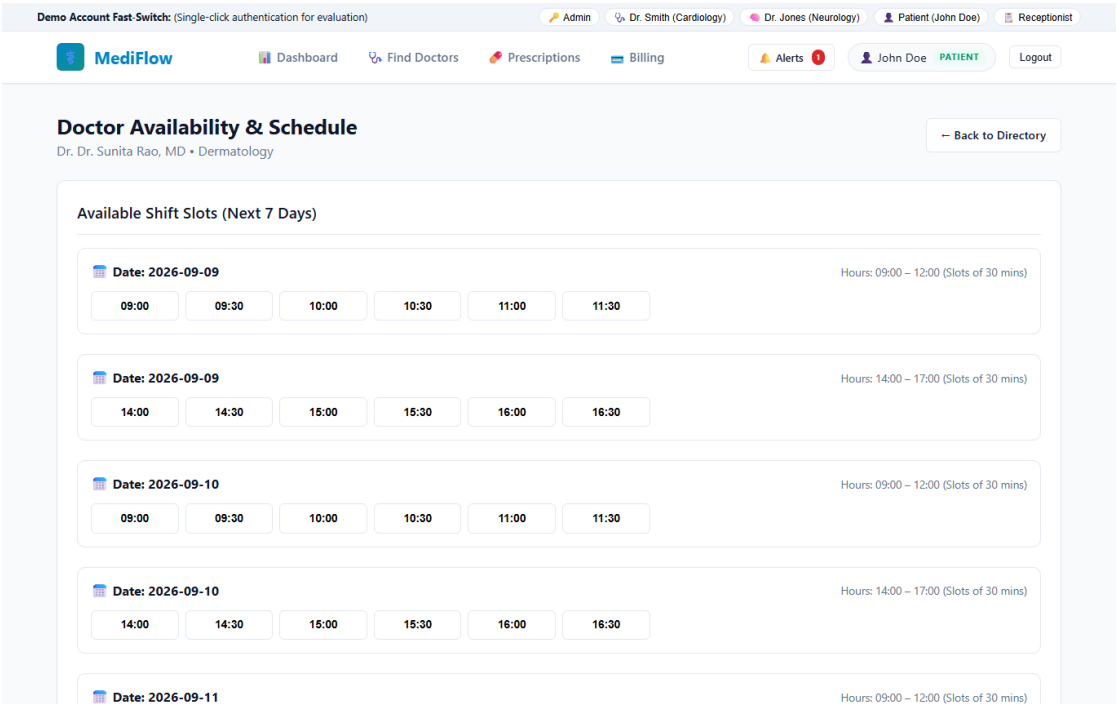


Figure 4: Doctor Availability Schedule & Slot Booking Interface

8.5 Billing & Invoices Management

Itemized breakdown of consultation fees and diagnostics with status badges and simulated one-click online settlement.

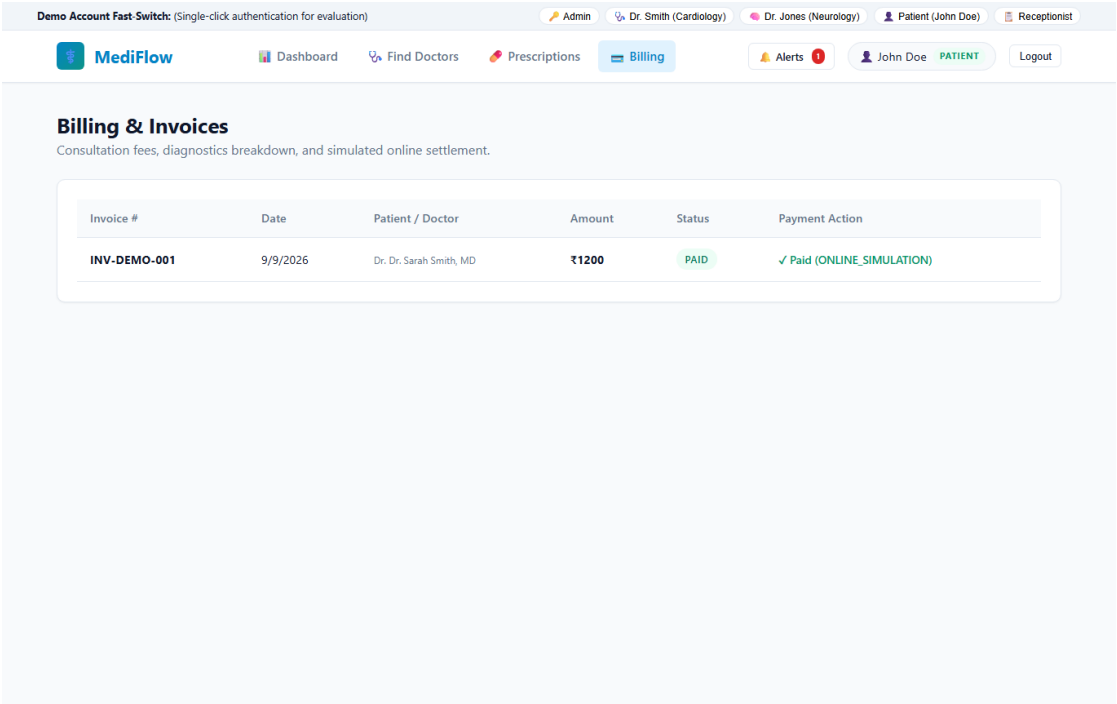


Figure 5: Patient Itemized Billing & Simulated Payment Interface

8.6 Digital Prescription Records

Chronological list of issued prescriptions with clinical diagnoses, attending physician details, and printable Rx summaries.

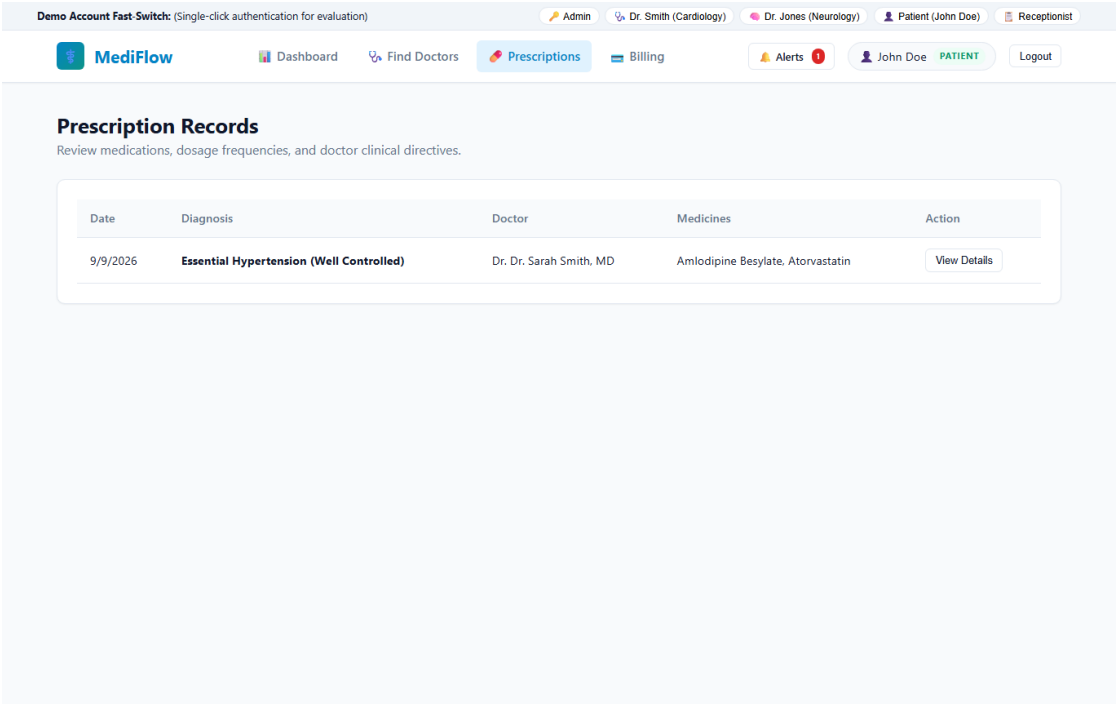


Figure 6: Digital Prescription Management & Medication Directive Records

8.7 Doctor Clinical Portal (Dr. Sarah Smith)

Enables attending physicians to manage assigned appointment queues, confirm visits, record clinical encounter notes, and issue digital prescriptions.

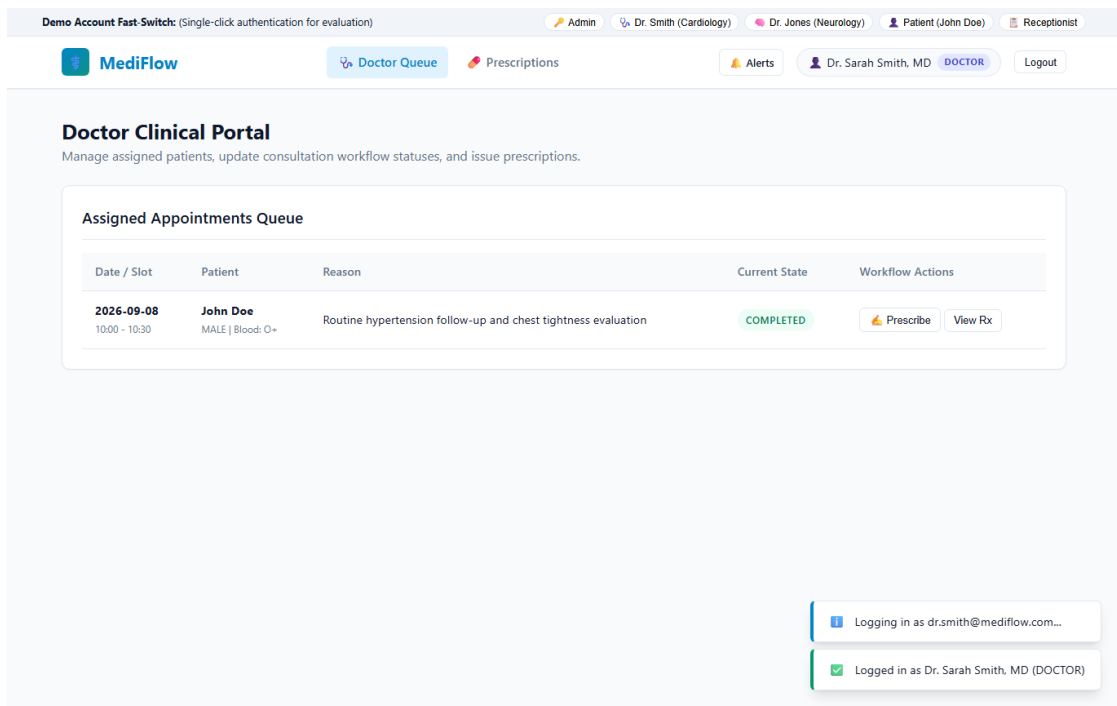


Figure 7: Doctor Clinical Workflow Portal & Assigned Queue

8.8 Hospital Executive Dashboard & Operational Reports

Aggregated KPI cards displaying total patients, active doctors, scheduled visits, and settled revenue in ₹ alongside appointment status distributions.

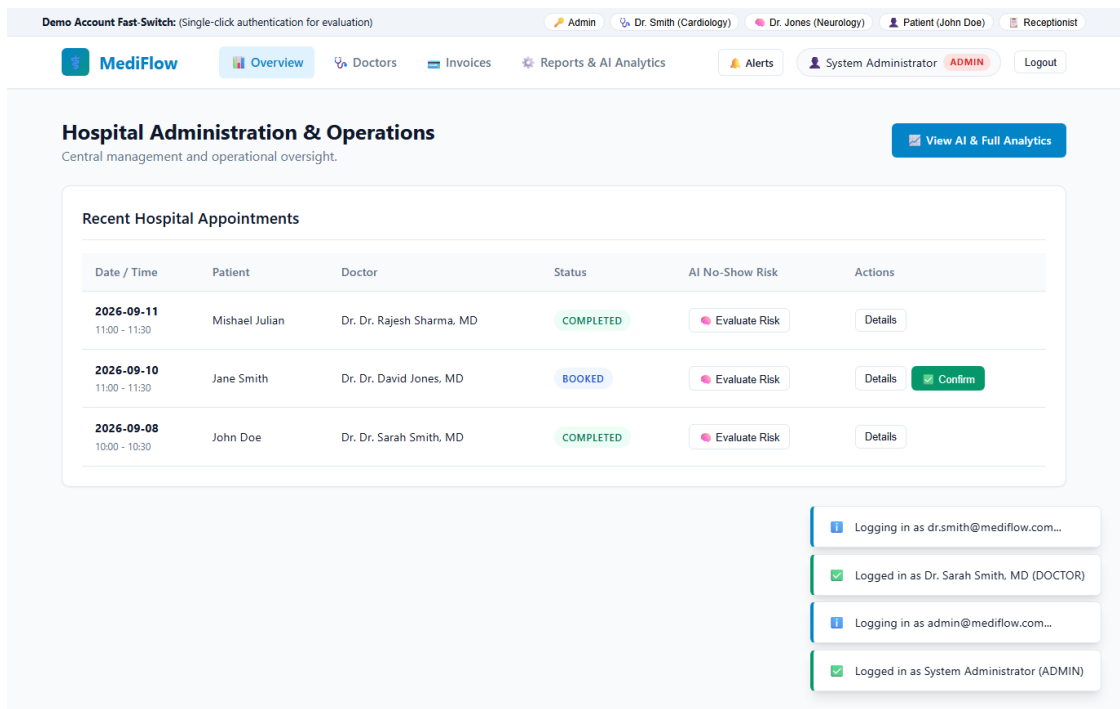


Figure 8: Administrator Executive KPI Overview & Operational Reports

8.9 AI Appointment No-Show Risk Engine

Operational predictive assessment evaluating appointment attendance probability based on booking lead times and historical cancellation ratios.

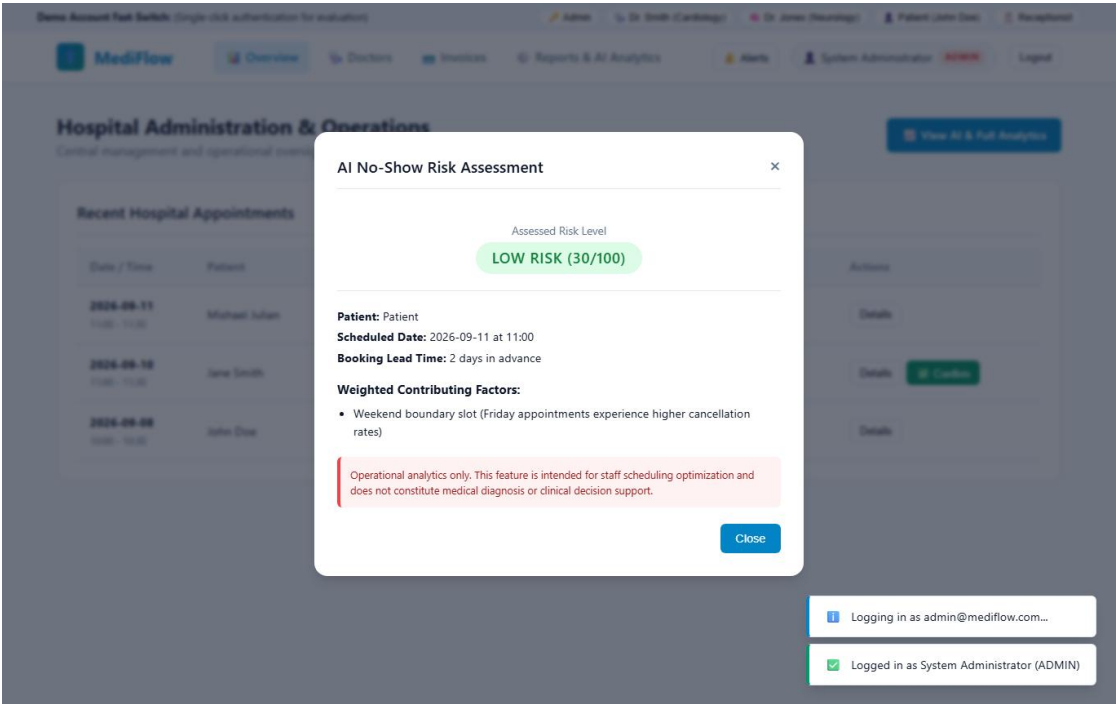


Figure 9: AI Appointment No-Show Risk Predictive Assessment Modal

9. Demo Accounts & Evaluation Walkthrough

The web portal (<http://localhost:5000>) includes a **1-Click Demo Fast-Switcher** bar for instant evaluator testing:

Demo Persona	Email	Password	Role & Test Purpose
System Administrator	admin@mediflow.com	Admin@123	Executive KPI Dashboard, Reports, User Controls, AI Risk Engine
Central Receptionist	receptionist@mediflow.com	Recep@123	Hospital Appointment Queue, Check-Ins, Inline Confirmations, Billing
Dr. Sarah Smith (Cardiology)	dr.smith@mediflow.com	Doctor@123	Assigned Consultation Queue, Clinical Notes, Prescription Issuance
Dr. David Jones (Neurology)	dr.jones@mediflow.com	Doctor@123	Neurology Patient Queue, Shift Schedule Management

John Doe (Active Patient)	john.doe@example.com	Patient@123	Completed Consultation, Active Prescription, Paid ₹1,200 Invoice
Jane Smith (Upcoming Patient)	jane.smith@example.com	Patient@123	Upcoming Neurology Appointment (BOOKED), Pending ₹900 Invoice

Recommended 5-Minute Evaluator Journey:

- 10. Patient Booking:** Log in as Patient → Browse Doctors → View 7-day shifts → Book an appointment → Observe immediate ₹ consultation invoice generation.
- 11. Conflict Test:** Attempt booking the exact same time slot with another account → Observe server 409 Conflict rejection.
- 12. Doctor Clinical Workflow:** Fast-switch to Doctor → Confirm appointment → Mark Complete with clinical diagnosis → Issue digital prescription.
- 13. Patient Review:** Fast-switch back to Patient → View newly issued prescription and pay pending invoice via simulation.
- 14. Admin Oversight:** Fast-switch to Admin → View updated revenue figures, department distribution charts, and AI No-Show Risk assessment.

10. Viva Voce Key Defense Concepts

- 15. Why MongoDB?** Flexible document schema perfectly modeling hierarchical clinical encounters (`items[]`, `lineItems[]`) with high read-heavy performance.
- 16. Double-Booking Prevention:** Dual-layered defense: (a) In-memory interval comparison algorithm in service layer, and (b) Database-enforced atomic sparse unique index on `slotKey`.
- 17. Security & RBAC:** Passwords never stored in plaintext (salted bcrypt hash with cost factor 10); JWT bearer tokens with payload verification on every protected route.
- 18. Standardized Responses:** Centralized custom error middleware mapping domain errors (`NOT_FOUND`, `SLOT_CONFLICT`, `FORBIDDEN`) to clean, predictable JSON HTTP codes.

11. Conclusion

MediFlow successfully delivers a complete, fault-tolerant, role-based healthcare management platform meeting all CIA-3 evaluation criteria. The system demonstrates high code quality, robust automated test coverage, thoughtful database modeling, and an intuitive user experience.

Submitted for CIA-3 Evaluation • Department of ADSE • Christ University